

SOLUTION

ARTIFICIAL INTELLIGENCE – CSA17

Assessment Tool 3 – Dry Run Test

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Course Code: CSA1717

Course Name: Artificial Intelligence

QUESTION 1 – DRY RUN OF A* SEARCH

Aim

To perform a dry run of A* Search and find the optimal path from A to G.

Formula

$$f(n) = g(n) + h(n)$$

Heuristic Values

Node	$h(n)$
A	7
B	6
C	4
D	3
E	2
G	0

Dry Run

Iteration	Current Node	Open List	Closed List
1	A	B(8), C(8)	A

2	B	E(6), C(8), D(12)	A, B
3	E	C(8), D(12)	A, B, E
4	C	D(12)	A, B, E, C
5	D	G(11)	A, B, E, C, D
6	G	Empty	A, B, E, C, D, G

Final Answer

Optimal Path: $A \rightarrow B \rightarrow D \rightarrow G$

Total Path Cost: $2 + 7 + 2 = 11$

Result

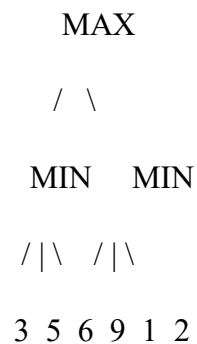
A* Search successfully found the optimal path from A to G with a minimum cost of **11**.

QUESTION 2 – DRY RUN OF MINIMAX WITH ALPHA-BETA PRUNING

Aim

To perform a dry run of Minimax with Alpha-Beta Pruning and determine the best move for MAX.

Game Tree



Dry Run

Initial:

$$\alpha = -\infty$$

$$\beta = +\infty$$

Step	Node	α	β	Value	Pruned
1	MAX	$-\infty$	$+\infty$	—	None
2	Left MIN	$-\infty$	3	3	None
3	MAX	3	$+\infty$	3	None
4	Right MIN	3	1	1	Node 2
5	MAX	3	$+\infty$	3	None

At the right MIN node:

$$\beta = 1$$

$$\alpha = 3$$

Since:

$$\beta \leq \alpha$$

$$1 \leq 3$$

the remaining node **2** is **pruned**.

Final Answer

Best Move for MAX: Left Subtree

Final Minimax Value: 3

Pruned Node: 2

Result

Minimax with Alpha-Beta Pruning successfully selected the left subtree with a final value of **3** while pruning the unnecessary node **2**.

CONCLUSION

A* Search found the optimal path **A** → **B** → **D** → **G** with cost **11**. Minimax with Alpha-Beta Pruning selected the **left subtree** with a final value of **3** and pruned node **2**. Both dry runs demonstrate efficient AI problem-solving and decision-making.